BIOMEDICAL LIMITED

ANNUAL PRODUCT QUALITY REVIEW

(YEAR – 2025)

Protocol No. : **BML/QA/APQR/QA/002**

Protocol Date : 23/04/2024

Total No. of Pages :

PRODUCT NAME : Dextrose 4.3% Saline Intravenous Injection

GENERIC NAME : Dextrose 4.3% Saline Intravenous Injection BP

COMPOSITION : Each 100ml contains

Dextrose BP	4.3gm
Sodium Chloride BP	0.18gm
Water for injection USP	q.s

PACK SIZE : 500ml in a printed PVC pouch

PACKAGING TYPES AVAILABLE : 500ml Polyvinylchloride PVC pouch
Corrugated cardboard boxes carton

MASTER FORMULAR NUMBER : BML/QA/MFR/007

SHELF LIFE: 3 Years

TIME PERIOD COVERED : JANUARY – DECEMBER, 2025

PRODUCT CODE : 4G

BIOMEDICAL LIMITED

1, Ohimege Road, Industrial Estate, Gaa- Imam, Ilorin Kwara State Nigeria.

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Sr. No	Subject	Name	Designation	Sign/Date
1	Compiled by	Olawale Ogunleye	Quality Assurance Officer	
2	Reviewed by	Robiu Idowu M.	Production Pharmacist	
3	Approved by	Koroma Abigael A.	Quality Assurance Manager	
4	Authorized by	Agboola L. A.	Factory Manager	

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1.0 SCOPE:

This covers internal annual products quality review, and all these parameters are monitored; changes in physical appearance of the product, chemical analysis (pH, Assay) and microbiological analysis (Sterility and Pyrogen), post market surveillance.

2.0 DESCRIPTION OF PRODUCT

Dextrose 4.3% Saline Intravenous Injection is a clear colorless sterile solution in a PVC pouch with double port for easy administration. Formulated for fluid replacement, treating electrolyte imbalance while providing calories, the clear solution's manufacturing process is carefully carried out to ensure the appropriate pH of the solution and a label claim of 4.3gm/100ml for Dextrose and 0.18gm/100ml for Sodium Chloride throughout the shelf-life is achieved. Careful control of all other parameters as mentioned in the scope and under the section for the review of the quality control of the product in this piece is assured at all times.

3.0 REVIEWS

3.1 REVIEW OF PRODUCT BATCHES MANUFACTURED

A batch of Dextrose 4.3% Saline Intravenous Injection comprises of about 3,050 bags. Total No. of Batches produced within the period January 2025 to December 2025 is 210 batches i.e. from Batch no.:4G25001 to 4G25210; all the batches produced have shown reliability of the manufacturing and the testing procedures in delivering constant and reproducible good product quality.

3.2 REVIEW OF STARTING AND PRIMARY PACKAGING MATERIALS USED IN THE FPP

The starting materials for the manufacture of Dextrose 4.3% Saline Intravenous Injection are Dextrose Monohydrate (active ingredient) Sodium Chloride (active ingredient) and Water for injection (excipient). Dextrose Monohydrate and Sodium Chloride are sourced from approved vendors while water for injection is produced in house. The starting materials are each tested to comply with respective pharmacopeia and in-house specifications. The table below shows the different consignment of Dextrose and Sodium Chloride received and used in the manufacture of Dextrose 4.3% Saline Intravenous Injection within the year under review. All the batches of the active material received met the specifications of pharmacopeia.

3.2.1 VENDORS

All the raw materials used in the production of these batches are sourced from the approved vendors only.

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3.2.2 RAW MATERIAL (API) MANUFACTURER/ SUPPLIER PERFORMANCE REVIEW

Name of API	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignment / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
Dextrose Monohydrate	Weifang Shengtai Medicine Co. Ltd. China	Weifang Shengtai Medicine Co. Ltd. China	**7**	**7**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Bdv Behrens GmbH Hamburg, Germany.	**1**	**1**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Kuber, Fzco Dubai Silicon, Oasis.	**2**	**2**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Farmasino Co.Ltd China.	**2**	**2**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Bdv Behrens GmbH Hamburg, Germany.	**2**	**2**	**Nil**	**Nil**
Sodium Chloride	K+S Minerals & Agriculture.	Fzco Dubai Silicon, Oasis.	**1**	**1**	**Nil**	**Nil**
	K+S Minerals & Agriculture.	K+S Minerals & Agriculture.	**1**	**1**	**Nil**	**Nil**
	Hebei Huachen Pharmaceutical Co. Ltd, China.	Hebei Huachen Pharmaceutical Co. Ltd, China.	**2**	**2**	**Nil**	**Nil**
	Salonen, Austria.	Muab Chemicals, Lagos.	**1**	**1**	**Nil**	**Nil**
	K+S Minerals & Agriculture.	Bdv, Behrens.	**1**	**1**	**Nil**	**Nil**

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3.2.3 PRIMARY PACKAGING MATERIAL

The FPP is contained in poly vinyl chloride (PVC) self-collapsible soft bag (pouch) bought as a finished product from the manufacturer of the PVC pouch. Manufacturer COA of the PVC is reviewed and testing done in-house to confirm the quality specifications stated by the manufacturers as conform to in-house specifications set for the primary packaging material.

Name of Primary packaging Material	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignments / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
PVC pouches	Jiangsu Medwell Medical Equipment Co. Ltd. China	Jiangsu Medwell Medical Equipment Co. Ltd. China	15	15	Nil	Nil
	Shanghai Kohope Medical Devices, Limited, China.	Shanghai Kohope Medical Devices, Limited, China.	3	3	Nil	Nil

3.3 REVIEW AND STATISTICAL ANALYSIS OF QUALITY CONTROL AND IN-PROCESS CONTROL

3.3.1 SPECIFICATIONS

Appearance: Clear, colorless and particles free solution
Dextrose B.P: 95-105%
Sodium Chloride B.P.: 95 – 105%
pH: 3.20 – 6.50

3.3.2 IN-PROCESS RESULTS:

The summary of in-process results of all the batches manufactured was prepared and the table showing the results presented. The summary of in-process results for the chemical analysis and microbiological analysis of the batches manufactured is shown

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Chemical Analysis of In-process product Result

S/N	BATCH NO	Analysis			
		Sodium Chloride Assay (95 – 105%)	Dextrose Assay (95 – 105%)	pH (sample) (3.2 – 6.50)	pH (WFI) (5.0 – 7.0)
1	4G25001	98.98	96.8	5.57	6.06
2	4G25002	95.78	98.52	6.22	6.49
3	4G25003	96.9	98.23	5.45	6.27
4	4G25004	96.9	98.8	5.32	6.2
5	4G25005	98.52	98.69	5.23	6.15
6	4G25006	96.9	96.09	5.85	6.2
7	4G25007	96.9	96.47	5.91	6.3
8	4G25008	98.04	98.14	5.71	6.2
9	4G25009	98.04	97.02	5.62	6.25
10	4G25010	96.9	98.98	5.93	6.24
11	4G25011	96.9	98.54	6.01	6.22
12	4G25012	96.9	98.49	6.22	6.06
13	4G25013	96.9	98.43	5.73	6.09
14	4G25014	96.9	98.52	6.09	6.11
15	4G25015	96.9	97.41	5.68	6.33
16	4G25016	98.04	97.9	5.66	6.27
17	4G25017	98.04	98.12	5.43	6.1
18	4G25018	96.41	97.59	5.58	6.19
19	4G25019	96.41	97.52	5.64	6.38
20	4G25020	98.04	97.26	5.55	6.14
21	4G25021	98.04	97.81	5.59	6.22
22	4G25022	98.04	97.83	5.9	6.1
23	4G25023	98.04	97.33	5.93	6.16
24	4G25024	96.41	96	5.65	6
25	4G25025	98.04	98.27	5.85	6.2
26	4G25026	96.41	96.29	6	6.12
27	4G25027	98.04	98.38	6.05	6.22
28	4G25028	98.04	96.82	5.8	6.21
29	4G25029	96.41	97.68	5.92	6.24
30	4G25030	98.04	97.26	5.88	6.27
31	4G25031	98.04	97.99	5.74	6.13
32	4G25032	98.04	98.58	5.9	6.13
33	4G25033	97.47	97.35	5.52	6.27
34	4G25034	97.47	97.48	5.48	6.32
35	4G25035	97.47	98.74	5.58	6.36
36	4G25036	97.47	96.84	5.7	6.05
37	4G25037	97.47	98.43	5.87	6.15
38	4G25038	95.84	97.24	5.8	6.13

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39	4G25039	97.47	97.88	5.73	6.2
40	4G25040	97.47	97.44	5.77	6.17
41	4G25041	95.84	97.64	5.9	6.21
42	4G25042	97.47	96.84	5.95	6.3
43	4G25043	96.9	98.83	5.99	6.27
44	4G25044	96.9	97.33	5.97	6.23
45	4G25045	96.9	98.14	5.9	6.23
46	4G25046	98.52	98.16	5.77	6.2
47	4G25047	96.9	97.28	5.9	6.25
48	4G25048	96.9	97.7	5.85	6.3
49	4G25049	96.9	98.6	5.65	6.1
50	4G25050	98.52	97.57	5.74	6.15
51	4G25051	96.9	98.34	5.73	6.17
52	4G25052	98.52	98.12	5.81	6.22
53	4G25053	96.9	97.15	5.97	6.2
54	4G25054	96.9	97.28	5.91	6.27
55	4G25055	96.9	97.5	5.93	6.3
56	4G25056	96.9	96.59	5.63	6.44
57	4G25057	96.34	98.27	5.37	6.33
58	4G25058	96.34	97.64	5.23	6.29
59	4G25059	96.34	98.16	5.36	6.21
60	4G25060	97.94	98.82	5.32	6.23
61	4G25061	96.34	98.54	5.49	6.36
62	4G25062	96.34	96.91	5.38	6.27
63	4G25063	97.94	98.16	5.82	6.4
64	4G25064	96.34	97.9	5.92	6.36
65	4G25065	96.34	97.15	5.47	5.85
66	4G25066	97.94	98.56	5.29	5.82
67	4G25067	96.34	98.74	5.24	5.91
68	4G25068	96.34	98.91	5.54	6.2
69	4G25069	97.94	98.47	5.76	6.33
70	4G25070	96.34	97.28	5.71	6.4
71	4G25071	96.34	98.21	5.47	6.38
72	4G25072	97.47	98.16	5.72	6.18
73	4G25073	97.47	98.43	5.78	6.15
74	4G25074	97.47	97.52	5.71	6.15
75	4G25075	97.47	97.08	5.76	6.23
76	4G25076	97.47	98.65	5.69	6.2
77	4G25077	97.47	98.23	5.8	6.23
78	4G25078	97.47	97.94	5.85	6.32
79	4G25079	97.47	98.74	5.78	6.3
80	4G25080	97.47	98.49	5.77	6.28
81	4G25081	97.47	97.26	6.01	6.38
82	4G25082	97.47	98.36	5.87	6.19
83	4G25083	97.47	98.52	5.82	6.16
84	4G25084	97.47	98.63	5.96	6.44

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85	4G25085	97.47	98.12	5.87	6.3
86	4G25086	96.9	98.41	5.89	6.12
87	4G25087	96.9	97.19	5.51	6.2
88	4G25088	96.9	97.86	6.17	6.31
89	4G25089	96.9	98.08	6.03	6.25
90	4G25090	96.9	98.52	6.17	6.27
91	4G25091	96.9	98.08	5.87	6.19
92	4G25092	96.9	98.74	6.18	6.35
93	4G25093	96.9	98.08	6.05	6.28
94	4G25094	97.31	97.33	5.83	6.23
95	4G25095	97.31	97.92	5.65	6.3
96	4G25096	98.87	97.33	5.54	6.15
97	4G25097	97.31	97.59	5.6	6.27
98	4G25098	97.31	97.68	5.44	6.07
99	4G25099	97.31	97.45	5.53	5.95
100	4G25100	98.87	97.52	5.84	6.09
101	4G25101	98.87	97.33	5.91	6.12
102	4G25102	97.31	97.86	5.96	6.28
103	4G25103	97.31	97.66	6.04	6.41
104	4G25104	97.31	97.78	5.88	6.04
105	4G25105	98.87	97.15	5.94	6.14
106	4G25106	98.87	97.37	6.02	6.32
107	4G25107	98.87	97.04	5.78	5.9
108	4G25108	97.31	97.19	5.9	6.13
109	4G25109	98.87	97.72	5.84	5.95
110	4G25110	97.31	98.03	5.76	6.03
111	4G25111	97.31	97.41	5.58	5.85
112	4G25112	97.31	97.61	5.87	6.29
113	4G25113	97.31	97.22	5.81	6.27
114	4G25114	97.31	97.06	5.76	6.21
115	4G25115	97.31	98.56	5.72	6.17
116	4G25116	97.31	96.84	5.64	6.11
117	4G25117	98.87	96.65	5.81	6.23
118	4G25118	97.31	97.46	5.7	6.14
119	4G25119	98.87	98.41	5.66	5.95
120	4G25120	97.31	96.47	5.47	5.83
121	4G25121	97.31	98.49	5.45	5.79
122	4G25122	97.31	98.01	5.49	5.81
123	4G25123	98.87	96.95	5.63	5.78
124	4G25124	97.31	97.88	5.58	5.7
125	4G25125	98.87	97.75	5.61	5.81
126	4G25126	97.31	98.41	5.72	5.9
127	4G25127	97.31	97.39	5.99	5.95
128	4G25128	98.87	98.49	6.31	6.41
129	4G25129	97.58	98.1	5.65	5.88
130	4G25130	97.58	96.99	5.67	5.91

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131	4G25131	96	97.26	6.16	6.52
132	4G25132	97.58	98.32	6.09	6.28
133	4G25133	97.58	98.6	6.25	6.36
134	4G25134	97.58	97.9	5.96	6.08
135	4G25135	96	97.77	5.88	6.16
136	4G25136	97.58	97.59	5.82	6.06
137	4G25137	96	98.25	5.66	5.85
138	4G25138	97.58	98.14	5.86	5.99
139	4G25139	96	96.18	5.95	6.33
140	4G25140	97.58	97.77	5.88	6.3
141	4G25141	97.58	98.03	6.01	6.34
142	4G25142	96	97.99	5.98	6.32
143	4G25143	97.58	97.72	6.05	6.29
144	4G25144	97.58	98.47	5.73	6.27
145	4G25145	97.58	98.32	5.85	6.3
146	4G25146	97.31	97.88	6.05	6.28
147	4G25147	97.31	97.44	5.87	6.16
148	4G25148	98.87	97.3	5.91	6.25
149	4G25149	97.31	97.59	5.85	6.12
150	4G25150	97.31	98.14	5.88	6.19
151	4G25151	98.87	97.22	5.94	6.23
152	4G25152	97.31	97.61	5.71	6.28
153	4G25153	97.31	97.99	5.68	6.17
154	4G25154	97.31	97.24	5.89	6.05
155	4G25155	98.98	98.96	5.61	5.9
156	4G25156	97.38	98.21	5.73	5.98
157	4G25157	98.98	97.37	5.85	6.26
158	4G25158	97.38	96.8	5.6	6.14
159	4G25159	97.38	97.72	5.91	6.31
160	4G25160	98.98	97.48	5.84	6.27
161	4G25161	98.98	97.37	5.95	6.33
162	4G25162	97.38	97.19	5.87	6.27
163	4G25163	98.98	97.79	5.89	6.3
164	4G25164	97.38	97.22	5.62	5.91
165	4G25165	97.38	98.14	5.4	5.7
166	4G25166	98.98	97.48	5.43	5.75
167	4G25167	97.38	98.38	5.48	5.81
168	4G25168	98.98	97.99	5.45	5.78
169	4G25169	98.98	98.45	5.57	5.94
170	4G25170	97.38	96.91	5.51	5.92
171	4G25171	97.38	98.23	5.98	6.28
172	4G25172	98.41	98.96	5.55	5.86
173	4G25173	98.41	98.85	5.61	5.92
174	4G25174	96.83	98.25	5.96	6.18
175	4G25175	98.41	97.64	5.8	6.09
176	4G25176	96.83	97.3	5.74	6.02

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177	4G25177	98.41	98.56	5.86	6.12
178	4G25178	96.83	97.46	5.79	6.04
179	4G25179	96.83	98.89	5.72	5.96
180	4G25180	98.41	97.99	5.75	5.98
181	4G25181	98.41	98.56	6.11	6.34
182	4G25182	98.41	98.83	6	6.21
183	4G25183	96.83	98.69	5.58	5.92
184	4G25184	96.83	98.01	5.64	5.87
185	4G25185	98.41	98.87	5.65	6.08
186	4G25186	98.41	98.95	5.58	6.02
187	4G25187	98.41	98.01	5.8	6.23
188	4G25188	96.83	98.47	5.97	6.3
189	4G25189	96.83	96.86	5.97	6.18
190	4G25190	98.41	98.96	5.87	6.29
191	4G25191	96.83	98.74	5.97	6.3
192	4G25192	98.41	97.19	5.95	6.27
193	4G25193	98.41	98.47	6.12	6.31
194	4G25194	97.38	97.88	5.41	5.86
195	4G25195	98.98	97.37	5.56	6.02
196	4G25196	97.38	97.7	5.66	6.04
197	4G25197	97.38	97.55	5.7	6.12
198	4G25198	98.41	97.22	5.59	5.91
199	4G25199	96.83	97.48	5.66	5.9
200	4G25200	98.41	97.57	5.78	6.12
201	4G25201	96.83	97.48	5.69	6.08
202	4G25202	96.83	97.75	5.72	6.1
203	4G25203	98.41	97.57	5.84	6.17
204	4G25204	98.41	97.44	5.61	6.11
205	4G25205	96.53	96.27	5.46	5.8
206	4G25206	98.41	97.92	5.48	5.84
207	4G25207	97.46	97.85	5.83	6.06
208	4G25208	96.73	97.85	5.85	6.33
209	4G25209	97.37	96.27	5.8	6.27
210	4G25210	96.69	97.85	5.91	6.09

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Chemical/Microbial Analysis for Finished product Result

S/N	BATCH NO	Analysis Results				
		Sodium Chloride Assay (95 – 105%)	Dextrose Assay (95 – 105%)	pH (sample) (4.50 – 7.00)	HMF (NMT 0.25)	Sterility (MUST BE STERILE)
1	4G25001	97.31	97.13	4.04	0.076	STERILE
2	4G25002	97.31	97.17	4.16	0.077	STERILE
3	4G25003	98.87	96.64	4.11	0.072	STERILE
4	4G25004	99.82	96.05	3.93	0.07	STERILE
5	4G25005	97.31	96.34	4.17	0.072	STERILE
6	4G25006	98.87	96.9	4.24	0.079	STERILE
7	4G25007	98.87	96.84	4.42	0.075	STERILE
8	4G25008	97.31	97.33	4.39	0.073	STERILE
9	4G25009	97.31	98.98	4.05	0.078	STERILE
10	4G25010	98.87	97.22	3.9	0.071	STERILE
11	4G25011	97.38	98.14	3.93	0.076	STERILE
12	4G25012	99.82	98.78	3.72	0.073	STERILE
13	4G25013	99.82	97.46	3.95	0.077	STERILE
14	4G25014	97.31	97.57	3.94	0.078	STERILE
15	4G25015	97.31	98.08	4.3	0.072	STERILE
16	4G25016	98.87	96.95	4.18	0.07	STERILE
17	4G25017	98.87	99.89	4.23	0.072	STERILE
18	4G25018	97.31	98.34	4.09	0.077	STERILE
19	4G25019	97.31	96.93	4.25	0.073	STERILE
20	4G25020	98.87	96.42	4.01	0.071	STERILE
21	4G25021	97.31	97.64	3.79	0.076	STERILE
22	4G25022	98.25	98.49	4.17	0.077	STERILE
23	4G25023	97.31	98.52	3.87	0.072	STERILE
24	4G25024	98.87	98.98	4.05	0.07	STERILE
25	4G25025	97.31	98.74	4.23	0.072	STERILE
26	4G25026	98.87	96.89	3.89	0.079	STERILE
27	4G25027	98.87	96.86	4.05	0.075	STERILE
28	4G25028	97.31	98.69	4.1	0.073	STERILE
29	4G25029	98.87	96.84	4.04	0.078	STERILE
30	4G25030	97.31	98.56	3.97	0.071	STERILE
31	4G25031	97.31	98.69	4.14	0.076	STERILE
32	4G25032	98.87	98.67	4.03	0.073	STERILE
33	4G25033	97.31	97.41	4.25	0.077	STERILE
34	4G25034	97.31	97.86	4.23	0.078	STERILE
35	4G25035	97.31	97.64	4.4	0.072	STERILE

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36	4G25036	97.58	97.41	4.19	0.07	STERILE
37	4G25037	96	98.08	4.18	0.072	STERILE
38	4G25038	97.58	98.74	4.3	0.077	STERILE
39	4G25039	97.58	98.3	4.26	0.073	STERILE
40	4G25040	97.58	98.52	4.12	0.071	STERILE
41	4G25041	96	97.86	4.27	0.076	STERILE
42	4G25042	96	97	4.39	0.077	STERILE
43	4G25043	97.58	96.43	4.27	0.072	STERILE
44	4G25044	96	97.13	3.97	0.07	STERILE
45	4G25045	97.58	96.73	4.08	0.072	STERILE
46	4G25046	97.58	97.08	4.13	0.079	STERILE
47	4G25047	96	97.55	3.9	0.075	STERILE
48	4G25048	96	98.83	4.26	0.073	STERILE
49	4G25049	97.58	98.34	4	0.078	STERILE
50	4G25050	96	97.72	4.17	0.071	STERILE
51	4G25051	97.58	97.79	4.11	0.076	STERILE
52	4G25052	97.58	98.25	4.05	0.073	STERILE
53	4G25053	98.87	97.64	4.01	0.077	STERILE
54	4G25054	98.87	98.72	4.2	0.078	STERILE
55	4G25055	98.87	97.55	4.1	0.072	STERILE
56	4G25056	97.31	98.87	4.01	0.07	STERILE
57	4G25057	97.31	98.36	4.02	0.072	STERILE
58	4G25058	97.31	97.41	3.83	0.077	STERILE
59	4G25059	98.87	98.94	3.88	0.073	STERILE
60	4G25060	97.31	97.81	4.01	0.071	STERILE
61	4G25061	97.31	97.33	3.87	0.076	STERILE
62	4G25062	97.38	98.52	3.83	0.077	STERILE
63	4G25063	98.98	97.22	3.8	0.072	STERILE
64	4G25064	98.98	97.37	3.93	0.07	STERILE
65	4G25065	98.98	98.89	4	0.072	STERILE
66	4G25066	97.38	97.33	4.12	0.079	STERILE
67	4G25067	98.98	98.16	4.07	0.075	STERILE
68	4G25068	97.38	97.52	3.94	0.073	STERILE
69	4G25069	98.98	97.23	4.02	0.078	STERILE
70	4G25070	97.38	97.19	4.05	0.071	STERILE
71	4G25071	98.98	97.04	3.9	0.076	STERILE
72	4G25072	97.38	98.6	3.84	0.073	STERILE
73	4G25073	98.98	97.02	3.72	0.077	STERILE
74	4G25074	98.98	98.03	3.92	0.078	STERILE
75	4G25075	97.38	98.54	4.02	0.072	STERILE
76	4G25076	97.38	97.75	3.8	0.07	STERILE
77	4G25077	97.38	97.08	3.91	0.072	STERILE
78	4G25078	98.98	98.45	3.84	0.077	STERILE
79	4G25079	96.82	98.12	3.95	0.073	STERILE
80	4G25080	96.82	97.28	3.98	0.071	STERILE
81	4G25081	98.41	96.97	3.77	0.076	STERILE

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82	4G25082	96.82	98.54	4.01	0.077	STERILE
83	4G25083	98.41	98.43	3.93	0.072	STERILE
84	4G25084	98.41	97.44	4.02	0.07	STERILE
85	4G25085	96.82	98.78	3.89	0.072	STERILE
86	4G25086	96.82	97.81	3.74	0.079	STERILE
87	4G25087	96.82	97.72	3.7	0.075	STERILE
88	4G25088	96.82	96.18	3.92	0.073	STERILE
89	4G25089	96.82	98.3	3.87	0.078	STERILE
90	4G25090	98.41	97.24	4	0.071	STERILE
91	4G25091	96.82	97.11	3.84	0.076	STERILE
92	4G25092	98.41	97.66	3.99	0.073	STERILE
93	4G25093	98.41	98.45	3.65	0.077	STERILE
94	4G25094	96.82	97.15	3.91	0.078	STERILE
95	4G25095	96.82	97.7	4.03	0.072	STERILE
96	4G25096	96.82	98.69	4.07	0.07	STERILE
97	4G25097	98.41	98.7	3.97	0.072	STERILE
98	4G25098	98.41	97.33	3.87	0.077	STERILE
99	4G25099	96.82	98.08	3.82	0.073	STERILE
100	4G25100	96.82	97.3	4.04	0.071	STERILE
101	4G25101	96.34	96.75	4.06	0.076	STERILE
102	4G25102	97.94	97.75	4.17	0.077	STERILE
103	4G25103	98.41	97.81	3.96	0.072	STERILE
104	4G25104	98.41	98.13	3.98	0.07	STERILE
105	4G25105	96.82	99.13	4.07	0.072	STERILE
106	4G25106	96.82	99.31	3.94	0.079	STERILE
107	4G25107	98.41	97.68	4.07	0.075	STERILE
108	4G25108	98.41	97.17	4.16	0.073	STERILE
109	4G25109	96.82	97.02	4.07	0.078	STERILE
110	4G25110	98.41	98.05	4.12	0.071	STERILE
111	4G25111	96.82	96.86	4.13	0.076	STERILE
112	4G25112	98.41	97.22	4.35	0.073	STERILE
113	4G25113	98.41	98.16	4.4	0.077	STERILE
114	4G25114	96.27	97.5	4.45	0.078	STERILE
115	4G25115	97.85	98.1	4.4	0.072	STERILE
116	4G25116	96.27	97.02	4.41	0.07	STERILE
117	4G25117	97.85	98.3	4.37	0.072	STERILE
118	4G25118	97.31	97.06	4.39	0.077	STERILE
119	4G25119	97.31	98.01	4.35	0.073	STERILE
120	4G25120	98.87	97.97	4.37	0.071	STERILE
121	4G25121	99.82	98.45	4.3	0.076	STERILE
122	4G25122	97.31	98.23	4.11	0.077	STERILE
123	4G25123	98.87	97.86	4.34	0.072	STERILE
124	4G25124	98.87	97.68	4.4	0.07	STERILE
125	4G25125	97.31	98.14	4.33	0.072	STERILE
126	4G25126	97.31	98.94	4.42	0.079	STERILE
127	4G25127	98.87	97.11	4.44	0.075	STERILE

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128	4G25128	97.38	98.87	4.47	0.073	STERILE
129	4G25129	99.82	97.17	3.98	0.078	STERILE
130	4G25130	99.82	97.13	4.08	0.071	STERILE
131	4G25131	97.31	97.68	4.22	0.076	STERILE
132	4G25132	97.31	98.49	4.36	0.073	STERILE
133	4G25133	98.87	98.32	4.41	0.077	STERILE
134	4G25134	98.87	98.38	4.26	0.078	STERILE
135	4G25135	97.31	96.8	4.28	0.072	STERILE
136	4G25136	97.31	96.89	4.17	0.07	STERILE
137	4G25137	98.87	97.83	4.4	0.072	STERILE
138	4G25138	97.31	97.17	4.03	0.077	STERILE
139	4G25139	98.25	98.49	4.41	0.073	STERILE
140	4G25140	97.31	96.97	4.12	0.071	STERILE
141	4G25141	98.87	98.01	4.18	0.076	STERILE
142	4G25142	97.31	96.71	4.22	0.077	STERILE
143	4G25143	98.87	96.53	4.2	0.072	STERILE
144	4G25144	98.87	96.67	4.22	0.07	STERILE
145	4G25145	97.31	98.16	4.08	0.072	STERILE
146	4G25146	98.87	96.4	4.26	0.079	STERILE
147	4G25147	97.31	97	4.36	0.075	STERILE
148	4G25148	97.31	97.77	4.07	0.073	STERILE
149	4G25149	98.87	97.35	4.25	0.078	STERILE
150	4G25150	97.31	98.05	4.1	0.071	STERILE
151	4G25151	97.31	97.55	4.12	0.076	STERILE
152	4G25152	97.31	97	4.09	0.073	STERILE
153	4G25153	97.58	98.78	4.3	0.077	STERILE
154	4G25154	96	97.68	4.02	0.078	STERILE
155	4G25155	97.58	98.87	4.36	0.072	STERILE
156	4G25156	97.58	98.23	4.37	0.07	STERILE
157	4G25157	97.58	98.85	4.2	0.072	STERILE
158	4G25158	96	97.66	4.02	0.077	STERILE
159	4G25159	96	97.39	4.01	0.073	STERILE
160	4G25160	97.58	97.83	4.01	0.071	STERILE
161	4G25161	96	98.36	4.22	0.076	STERILE
162	4G25162	97.58	98.52	4.18	0.077	STERILE
163	4G25163	97.58	98.78	4.1	0.072	STERILE
164	4G25164	96	98.52	4.25	0.07	STERILE
165	4G25165	96	97.02	4.12	0.072	STERILE
166	4G25166	97.58	97.94	4.06	0.079	STERILE
167	4G25167	96	98.3	4.22	0.075	STERILE
168	4G25168	97.58	97.68	4.17	0.073	STERILE
169	4G25169	97.58	97.61	3.97	0.078	STERILE
170	4G25170	98.87	97.59	3.86	0.071	STERILE
171	4G25171	98.87	97.81	4.14	0.076	STERILE
172	4G25172	98.87	97.7	4.27	0.073	STERILE
173	4G25173	97.31	98.72	4.22	0.077	STERILE

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174	4G25174	97.31	98.58	4.27	0.078	STERILE
175	4G25175	97.31	98.03	4.18	0.072	STERILE
176	4G25176	98.87	97.17	4.12	0.07	STERILE
177	4G25177	97.31	97.39	4.09	0.072	STERILE
178	4G25178	97.31	97.22	4.07	0.077	STERILE
179	4G25179	97.38	98.98	4.14	0.073	STERILE
180	4G25180	98.98	98.54	4.03	0.071	STERILE
181	4G25181	98.98	98.05	4.13	0.076	STERILE
182	4G25182	98.98	96.82	4.1	0.077	STERILE
183	4G25183	97.38	97.24	3.92	0.072	STERILE
184	4G25184	98.98	98.63	3.93	0.07	STERILE
185	4G25185	97.38	98.12	4.02	0.072	STERILE
186	4G25186	98.98	98.23	3.88	0.079	STERILE
187	4G25187	97.38	98.34	4	0.075	STERILE
188	4G25188	98.98	97.77	4.16	0.073	STERILE
189	4G25189	97.38	98.05	4.12	0.078	STERILE
190	4G25190	98.98	97.24	4.02	0.071	STERILE
191	4G25191	98.98	97.02	4.13	0.076	STERILE
192	4G25192	97.38	98.52	3.86	0.073	STERILE
193	4G25193	97.38	98.78	3.97	0.077	STERILE
194	4G25194	97.38	97.44	3.98	0.078	STERILE
195	4G25195	98.98	97.81	3.89	0.072	STERILE
196	4G25196	96.82	98.74	4.01	0.07	STERILE
197	4G25197	96.82	98.12	3.99	0.072	STERILE
198	4G25198	98.41	98.32	4.06	0.077	STERILE
199	4G25199	96.82	97.46	4.17	0.073	STERILE
200	4G25200	98.41	96.97	4	0.071	STERILE
201	4G25201	98.41	97.22	3.76	0.076	STERILE
202	4G25202	96.82	98.47	3.82	0.077	STERILE
203	4G25203	96.82	96.75	4	0.072	STERILE
204	4G25204	96.82	98.08	3.81	0.07	STERILE
205	4G25205	96.82	97.41	4.14	0.072	STERILE
206	4G25206	96.82	96.91	4.11	0.079	STERILE
207	4G25207	98.41	96.62	3.79	0.075	STERILE
208	4G25208	96.82	97.81	3.85	0.073	STERILE
209	4G25209	98.41	97.68	4.1	0.078	STERILE
210	4G25210	98.41	96.75	3.9	0.071	STERILE

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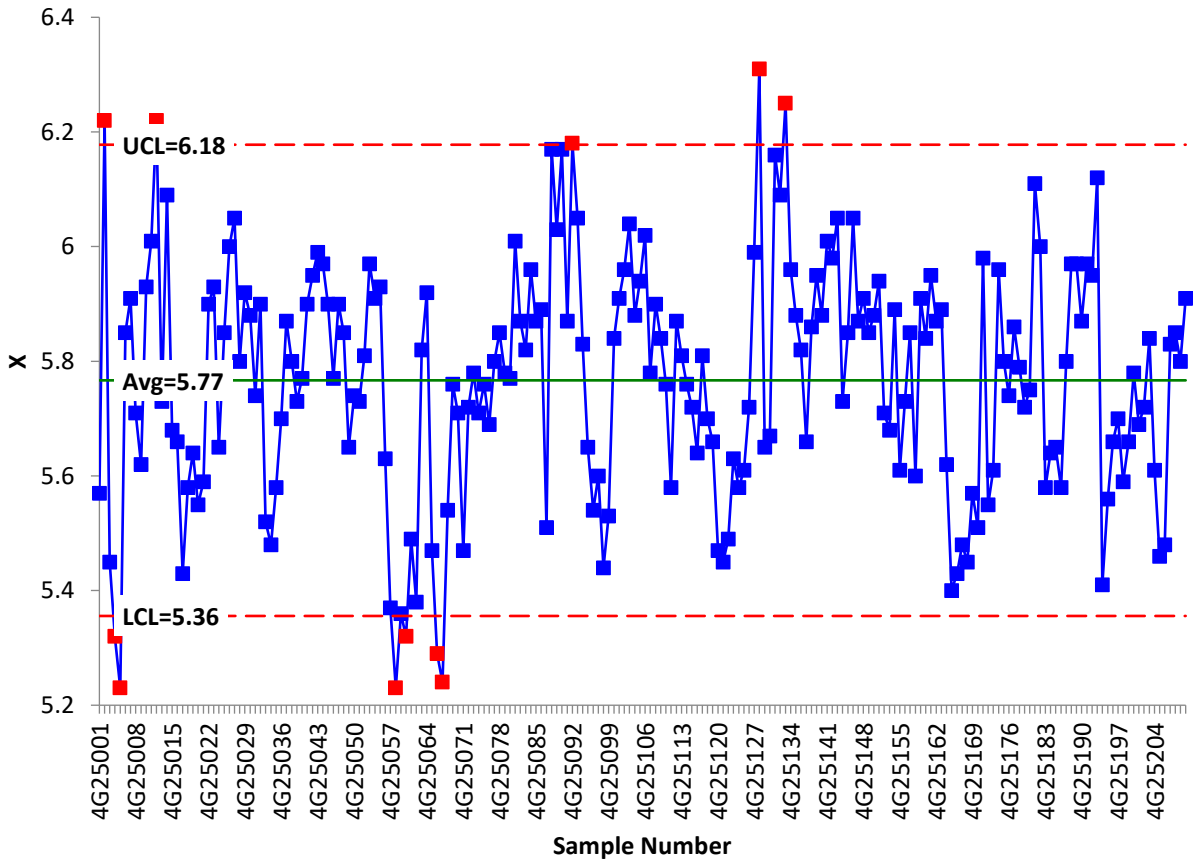
3.3.3 TREND ANALYSIS

Trend analysis of Dextrose content, Sodium Chloride content, HMF content, pH, Pyrogen and Final yield %, were carried out on all the above batches.

Annexure I: pH trend for inprocess analysis of the product

Specification (3.20-6.50**)**



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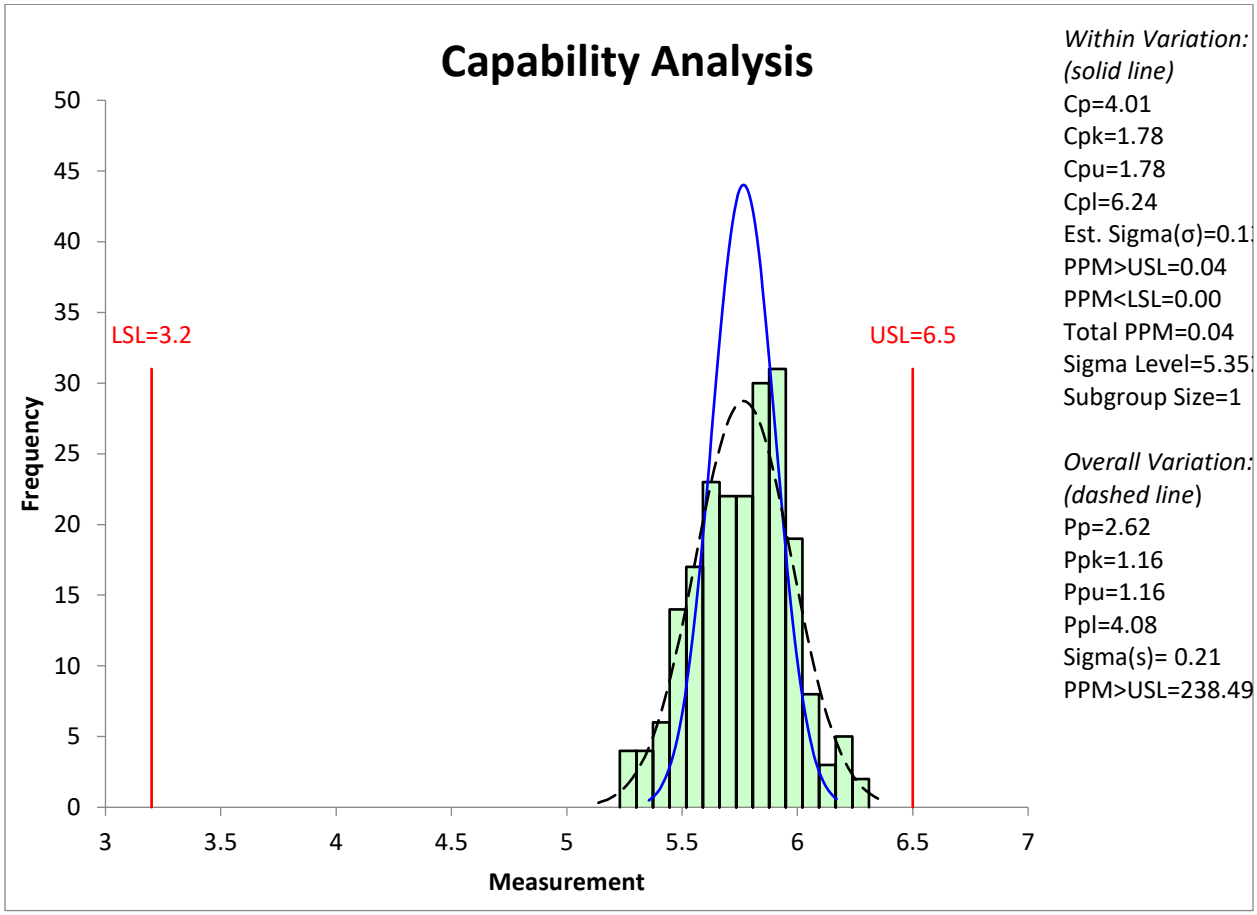

Mean	5.766714286
Standard deviation	0.209908078
Maximum	6.31
Minimum	5.23
Control limit: 5.14–6.40	
Alert limit: 5.35–6.19	
Minimum-Maximum: 5.23–6.31	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

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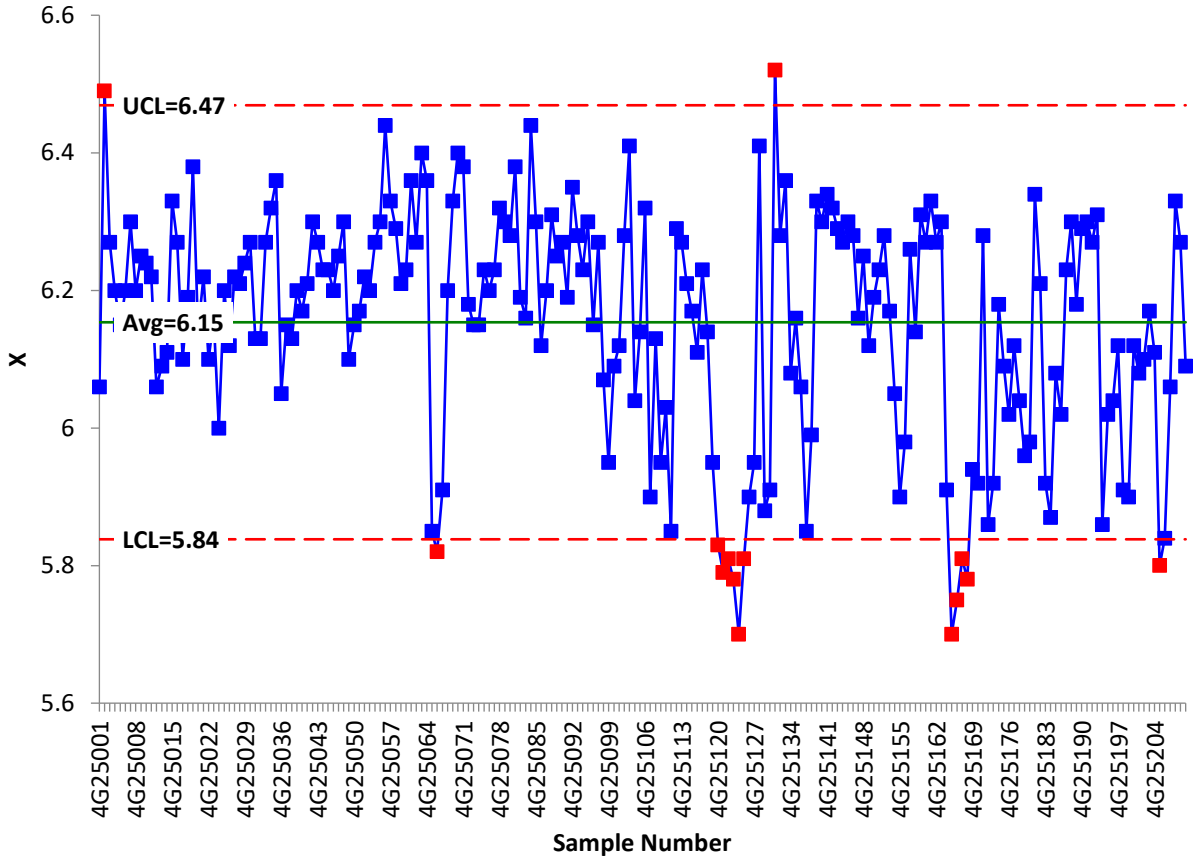
Annexure II: pH trend for Water for Injection (WFI)

Specification (5.0 -7.0**)**



CONTROL COPY

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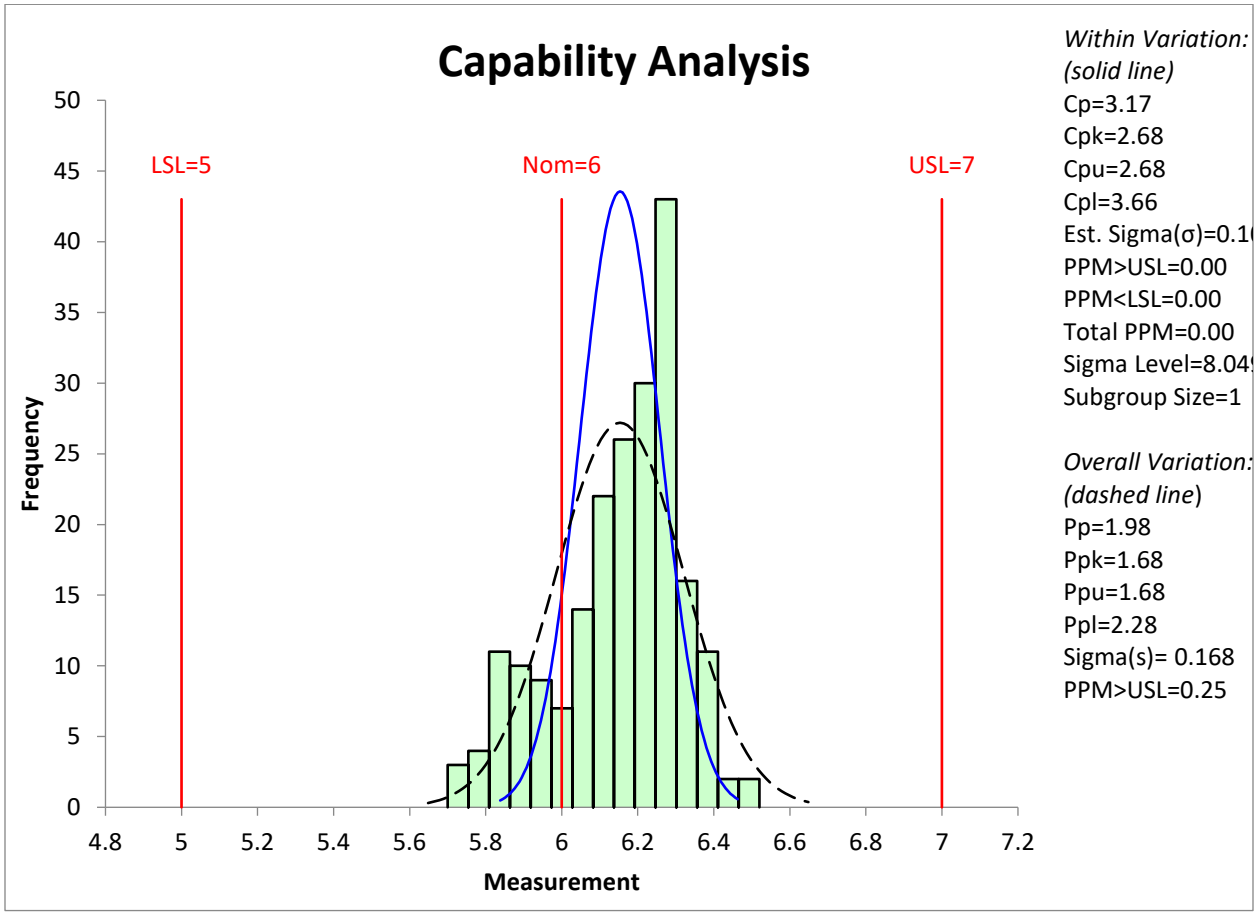

Mean	6.153667
Standard deviation	0.168402
Maximum	6.52
Minimum	5.7
Control limit: 5.65–6.66	
Alert limit: 5.82–6.49	
Minimum-Maximum: 5.70–6.52	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

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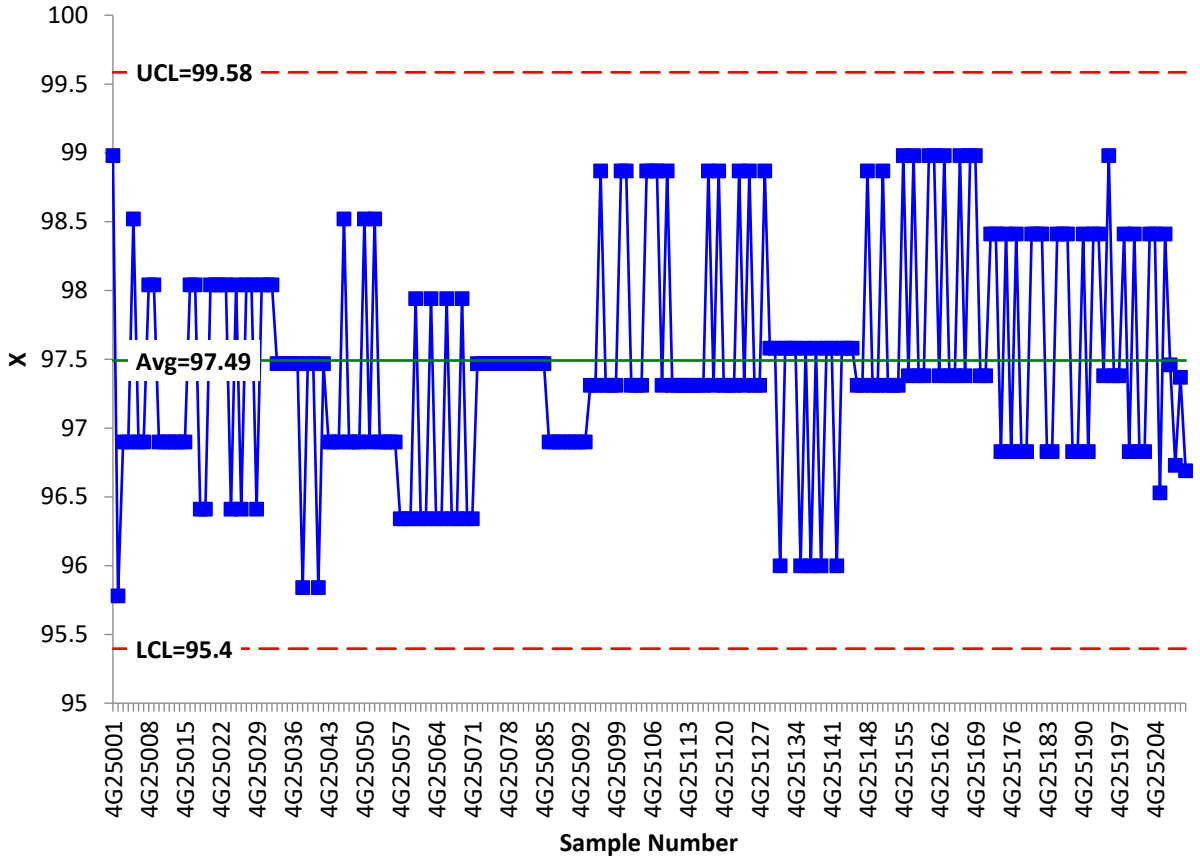
Annexure III: Assay trend of in-process analysis of Sodium Chloride in Dextrose 4.3% Saline Injection
Specification of Sodium Chloride (95 – 105%**)**



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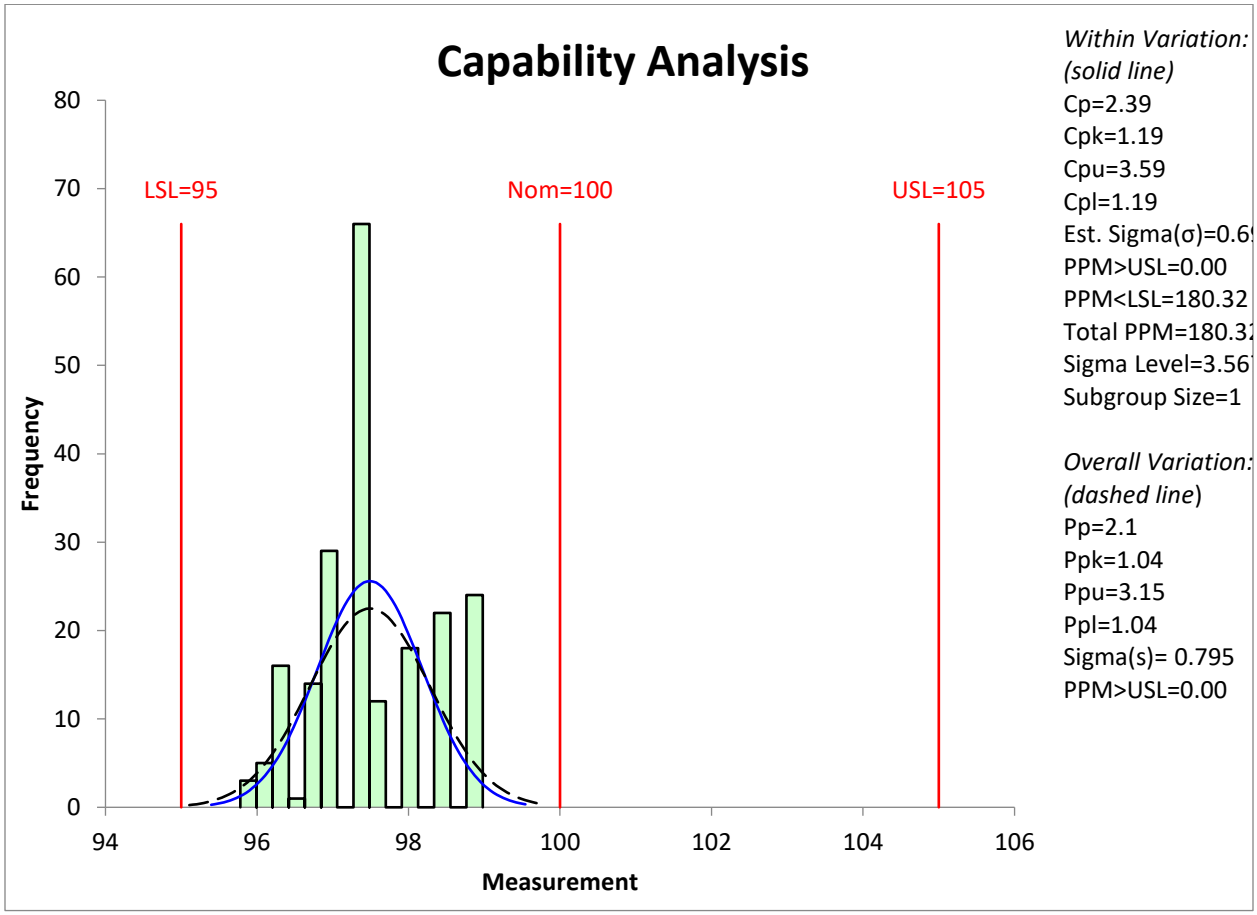

Mean	97.49052381
Standard deviation	0.79452445
Maximum	98.98
Minimum	95.78
Control limit: 95.11–99.87	
Alert limit: 95.90–99.08	
'Minimum-Maximum: 95.78–98.98	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

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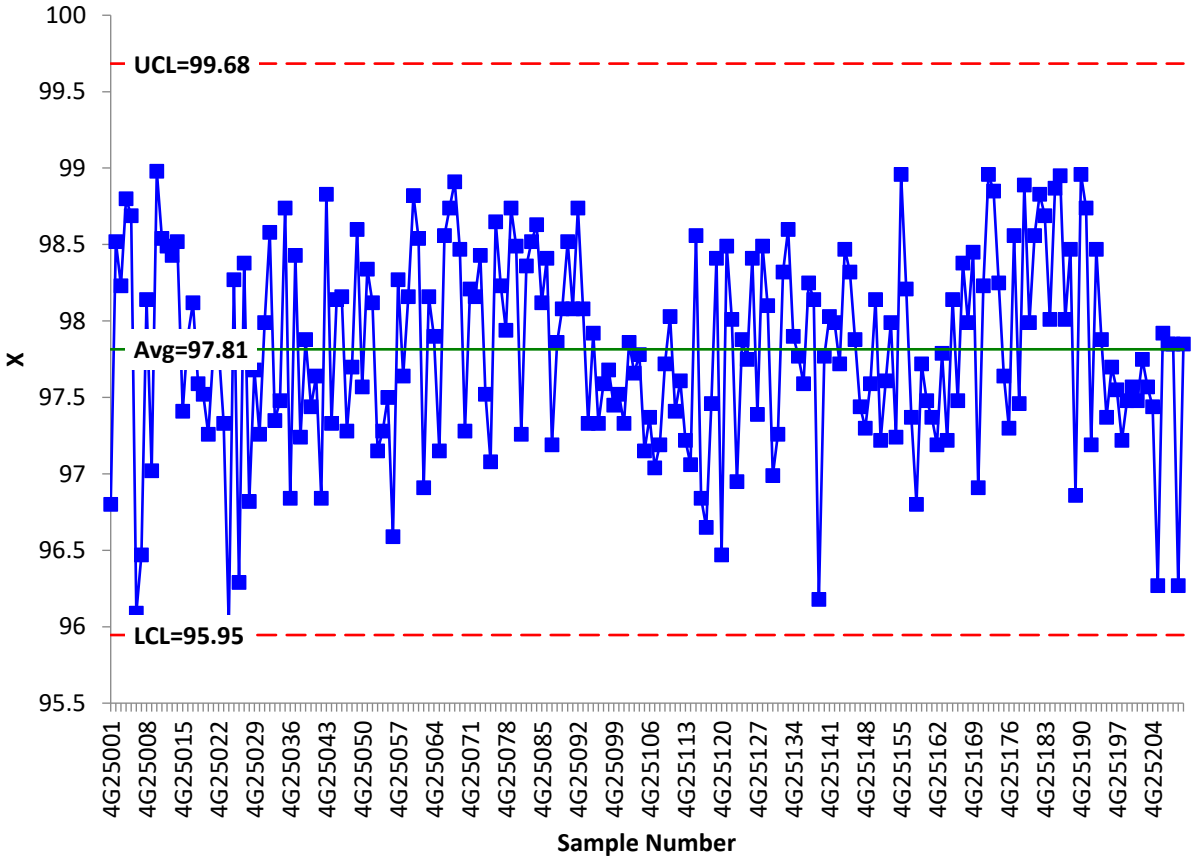
Annexure IV: Assay trend of in-process analysis of Dextrose in Dextrose 4.3% Saline Injection

Specification of Dextrose (95 – 105%**)**



CONTROL COPY

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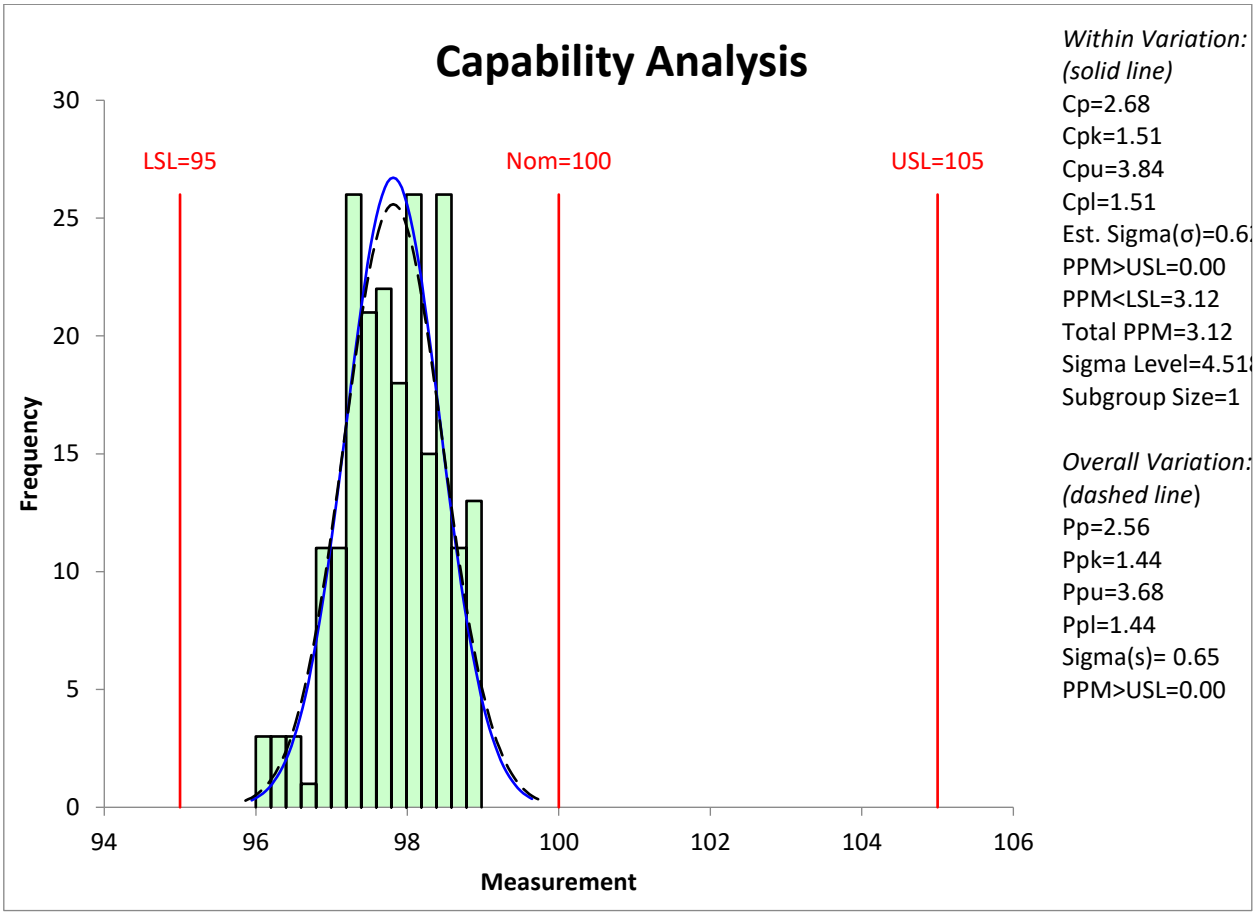

Mean	97.81466667
Standard deviation	0.650415497
Maximum	98.98
Minimum	96.00
Control limit: 95.86–99.77	
Alert limit: 96.51–99.12	
'Minimum-Maximum: 96.00–98.98	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

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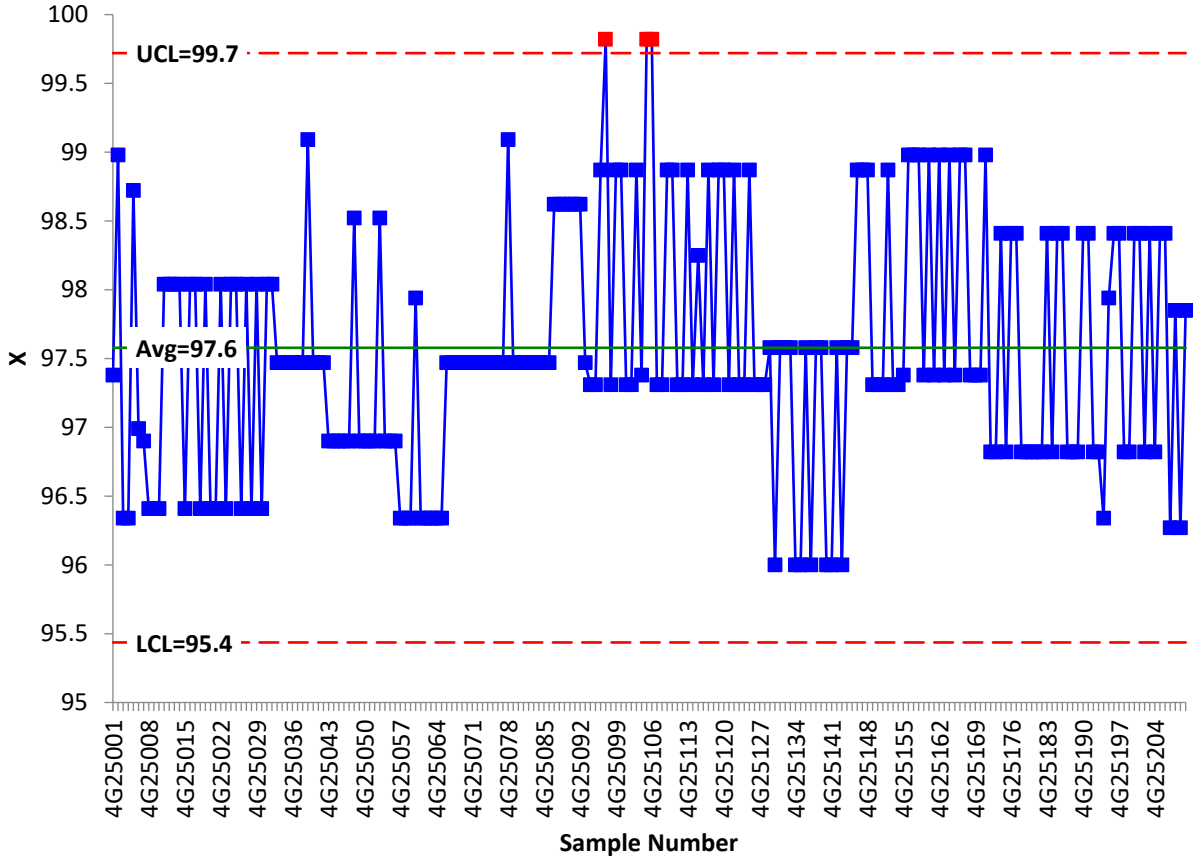
Annexure V: Assay trend of finished product analysis of Sodium Chloride in Dextrose 4.3% Saline Injection

Specification of Sodium Chloride (95 – 105%)



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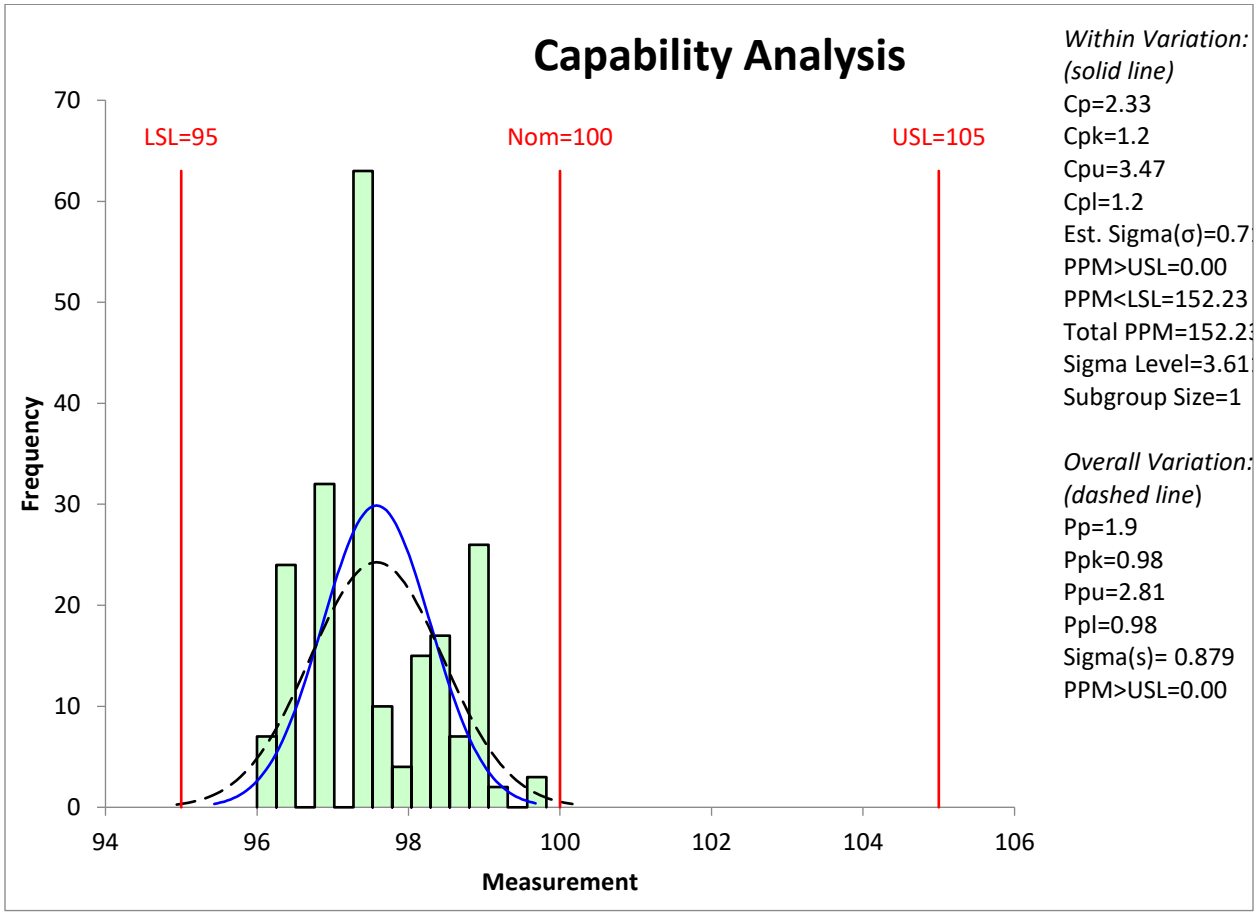

Mean	97.57833333
Standard deviation	0.879371559
Maximum	99.82
Minimum	96
Control limit: 94.94–100.22	
Alert limit: 95.82–99.34	
Minimum-Maximum: 96.00–99.82	
Alert limit within the specification limit, control limit existing within the specification limit.	

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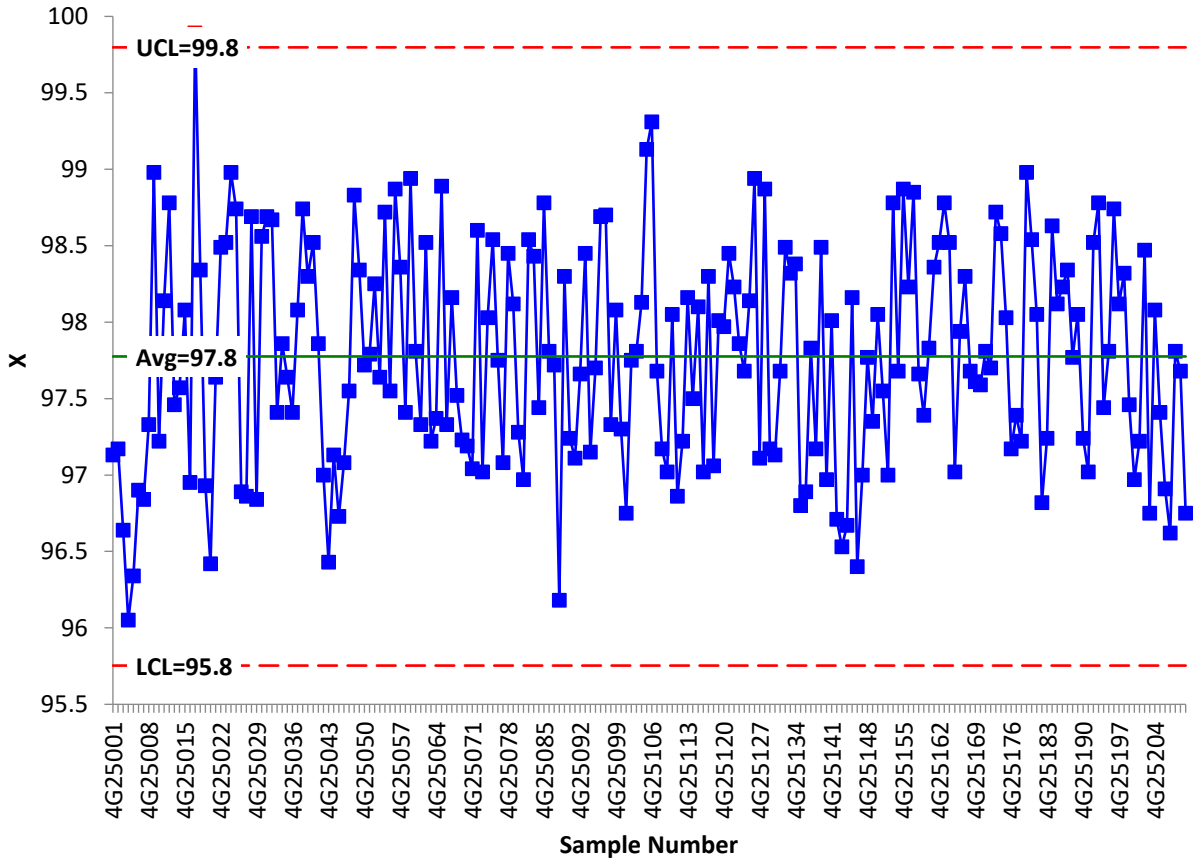
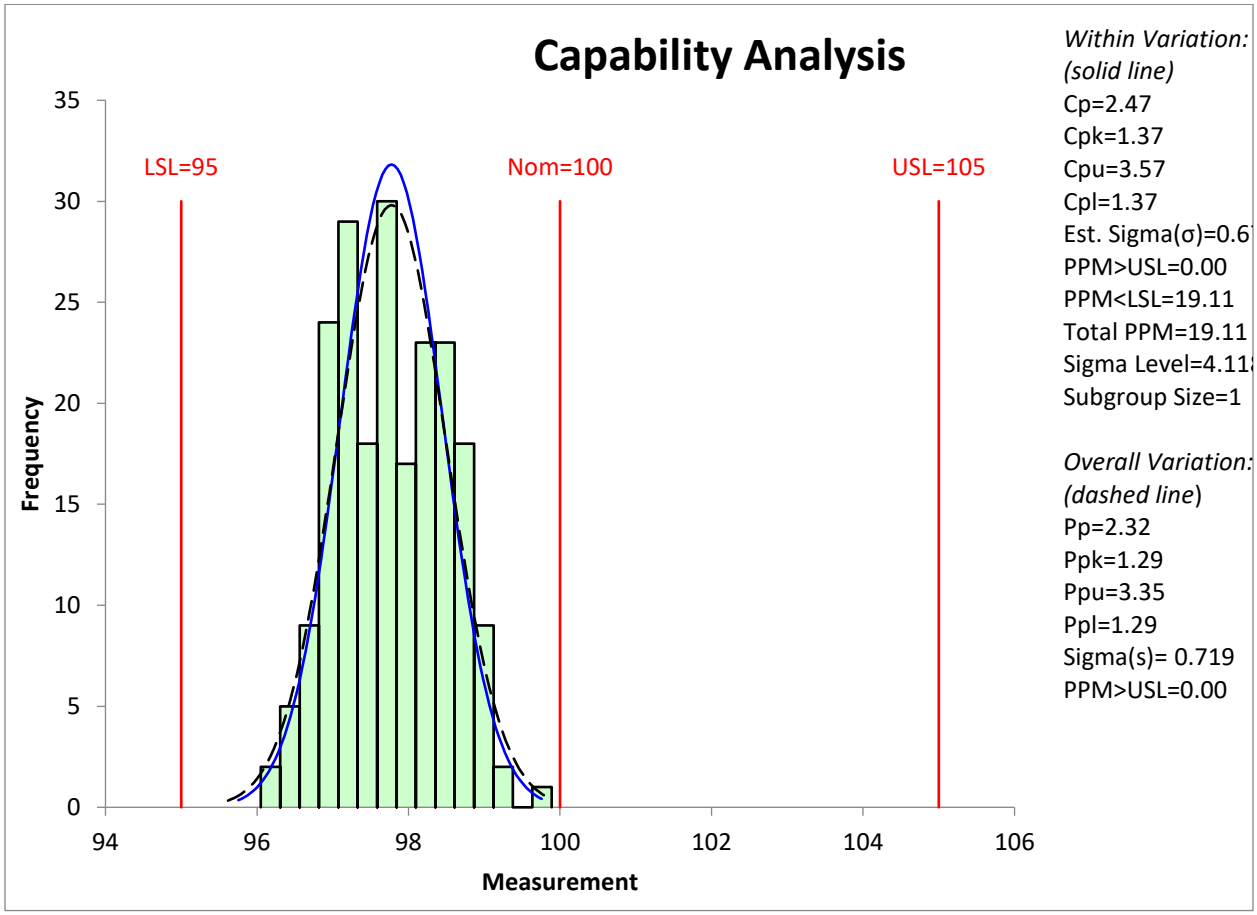
Annexure VI: Assay trend of finished product analysis of Dextrose in Dextrose 4.3% Saline Injection
Specification of Dextrose (95 – 105%)



CONTROL COPY

MASTER COPY



Mean	97.77557143
Standard deviation	0.719333761
Maximum	99.89
Minimum	96.05
Control limit: 95.62–99.93	
Alert limit: 96.34–99.21	
Minimum-Maximum: 96.05–99.89	
Alert limit within the specification limit, control limit existing within the specification limit.	

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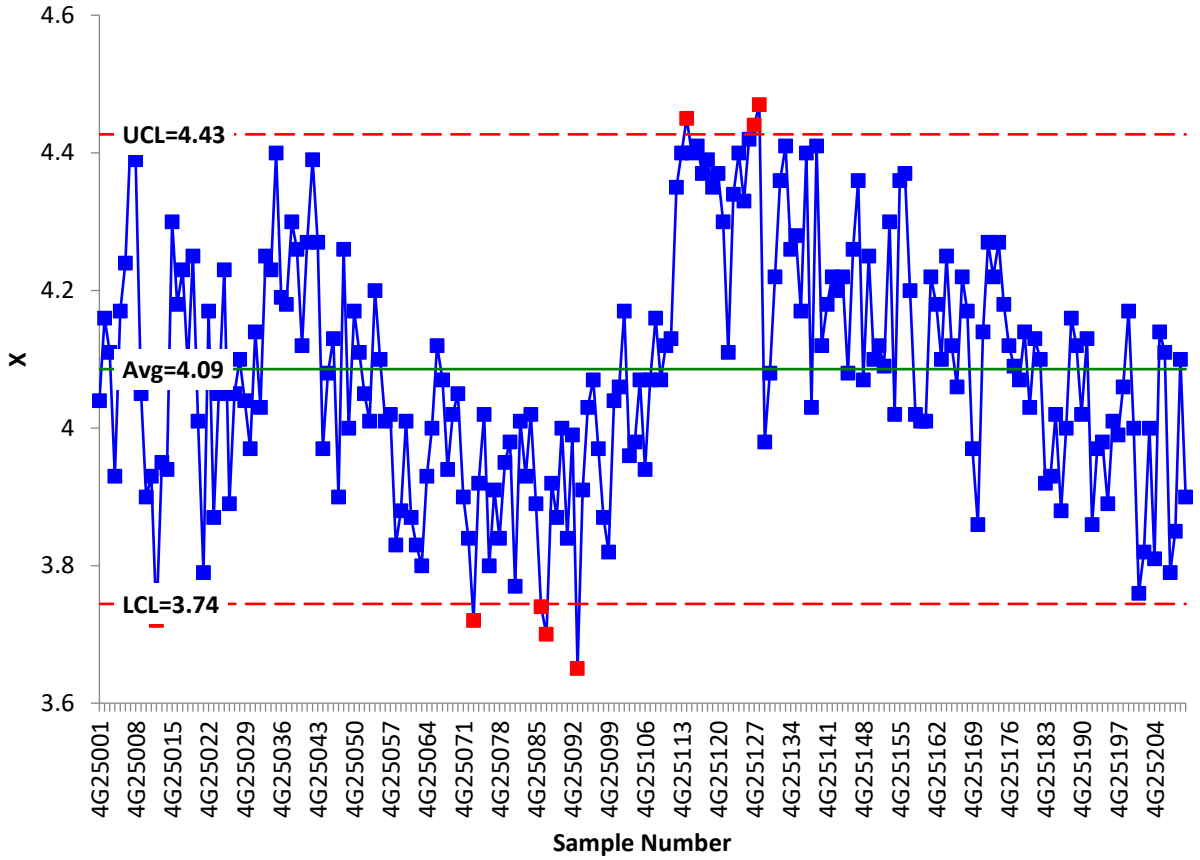
Annexure VII: pH trend of finished product analysis of Dextrose 4.3% Saline Injection

Specification of pH (3.20-6.50**)**



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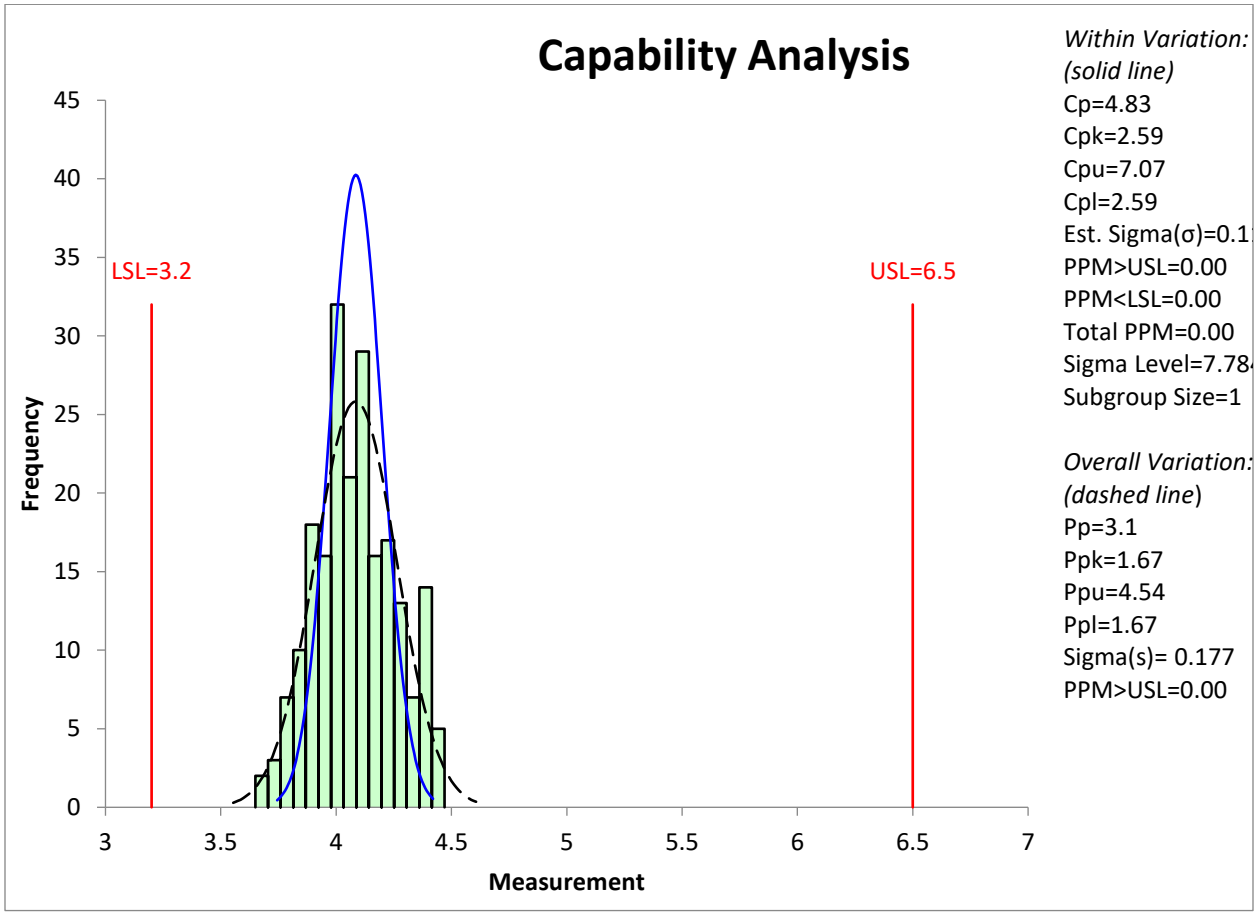

Mean	4.085571429
Standard deviation	0.177275789
Maximum	4.47
Minimum	3.65
Control limit: 3.55–4.62	
Alert limit: 3.70–4.44	
Minimum-Maximum: 3.65–4.47	
Alert limit within the specification limit, control limit existing within the specification limit.	

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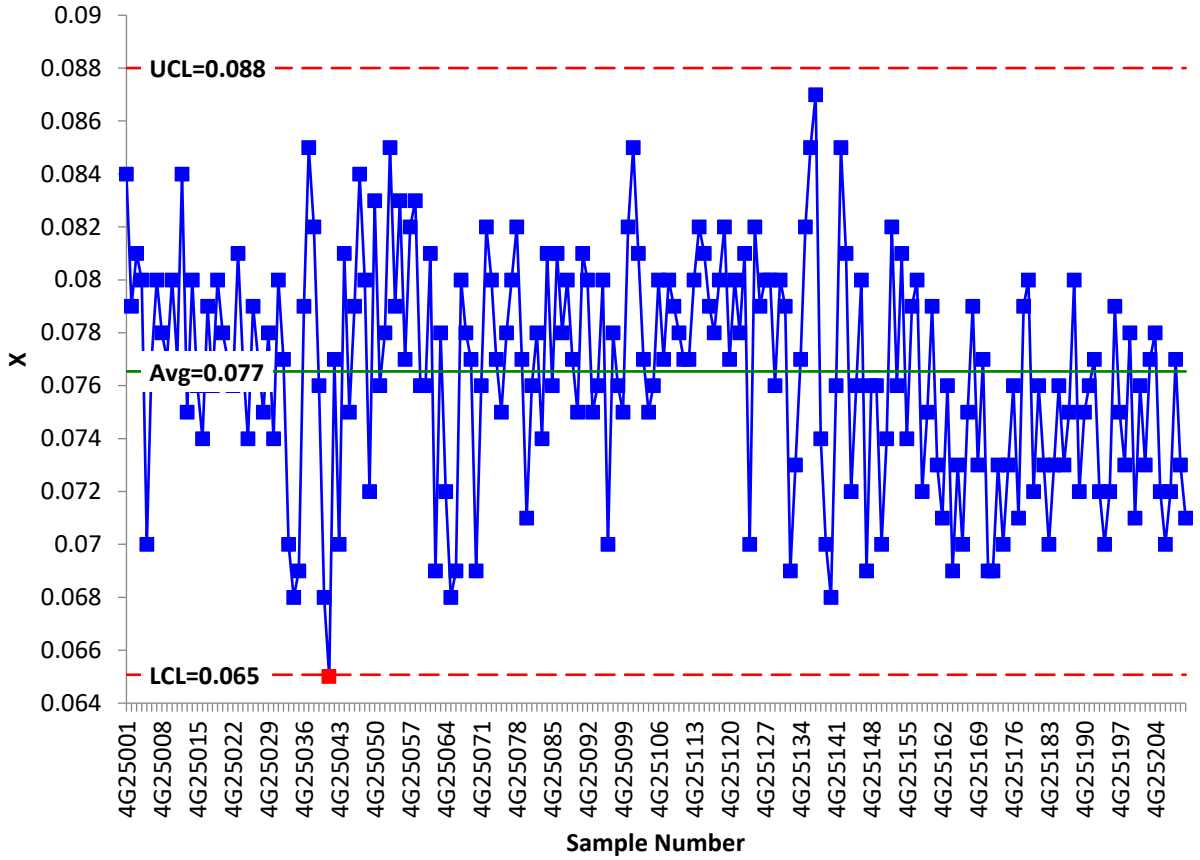
Annexure VIII: HMF trend of finished product analysis of Dextrose 4.3% Saline Injection

Specification of HMF (NMT 0.25**)**



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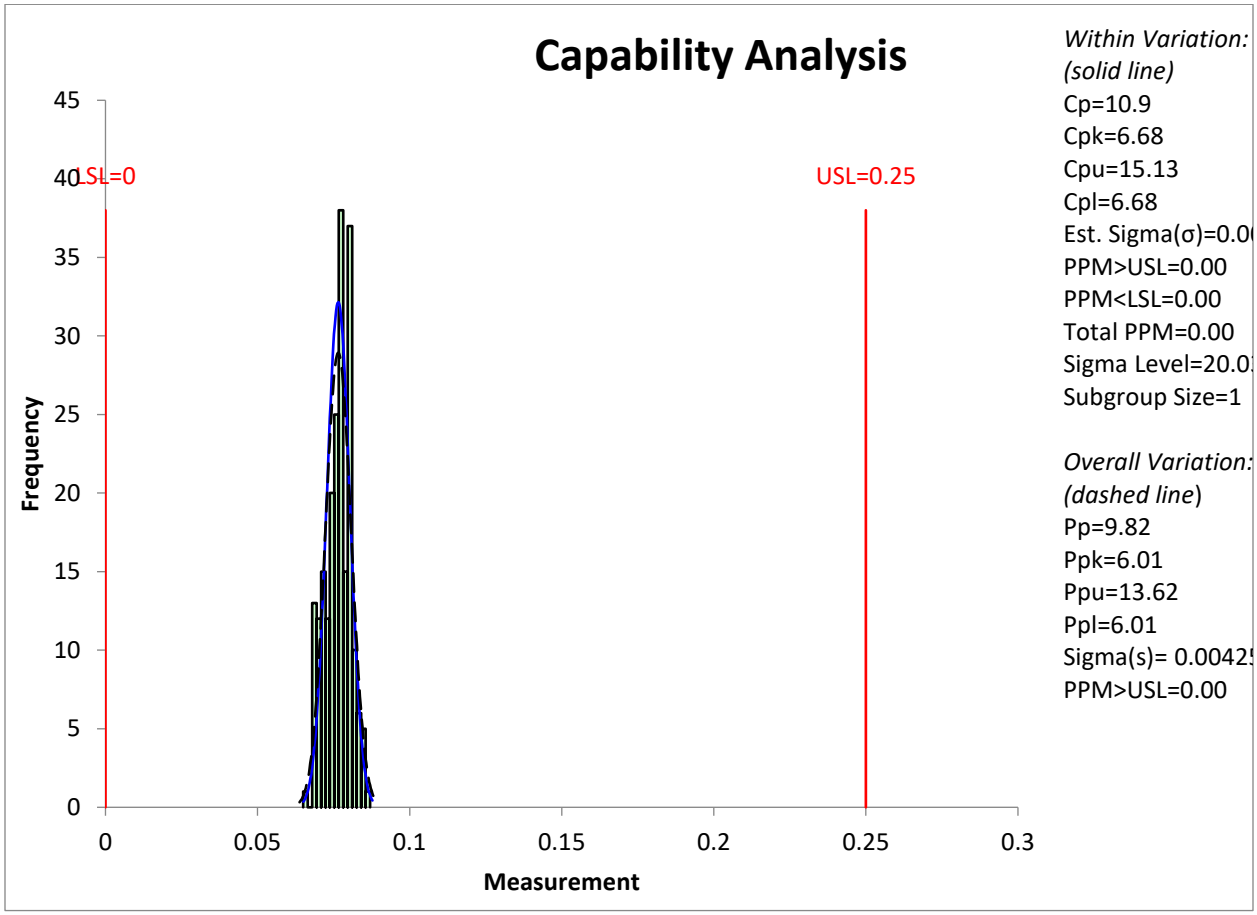

Mean	0.076538095
Standard deviation	0.004245006
Maximum	0.087
Minimum	0.065
Control limit: 0.064–0.089	
Alert limit: 0.068–0.085	
Minimum-Maximum: 0.065–0.087	
Alert limit within the specification limit, control limit existing within the specification limit.	

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SUMMARY OF PROCESS CAPABILITY (acceptance criteria ≥ 1.0)

PROCESS CAPABILITY FOR INTERMEDIATE		
1	pH	1.78
2	Rinse Water	2.68
3	Sodium Chloride	1.19
4	Dextrose	1.51

PROCESS CAPABILITY FOR PRODUCT		
1	pH	2.59
2	Sodium Chloride	1.20
3	Dextrose	1.37
4	HMF	6.68

CQAs	Intermediate		Finished product	
	Control limit	Process capability	Process capability	Control limit
pH	CL within SL	1.78	2.59	CL within SL
Rinse water	CL within SL	2.68	-	-
HMF	-	-	6.68	CL within SL
Sodium chloride content	CL within SL	1.19	1.20	CL within SL
Dextrose content	CL within SL	1.51	1.37	CL within SL

3.4 REVIEW OF BATCHES WITH OUT-OF-SPECIFICATION

No batch was recorded having out of specifications on any quality parameters.

3.5 REVIEW OF CRITICAL DEVIATIONS, NON-CONFORMANCES AND RELATED INVESTIGATIONS

3.5.1 PROCESS DEVIATIONS

No process deviation was reported from the standard processes for batches manufactured during the period Jan., 2025 to Dec., 2025.

3.5.2 YIELD DEVIATIONS

No observed deviation from the predetermined tolerance was observed.

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3.5.3 NON-CONFORMANCE

No observed non-conformance from the predetermined tolerance was observed.

3.5.4 BATCH FAILURES

There was no batch failure observed during the period Jan., 2025 to Dec., 2025.

3.6 CHANGE CONTROL

There was no change done to the processes, equipment or analytical methods for the manufacture of Dextrose4.3% Saline Intravenous Injection during the year under review. As such, no change control was recorded throughout the period Jan., 2025 to Dec., 2025.

3.6.1 REGULATORY CHANGE

No regulatory changes as well as Master Formula Changes were made concerning these products throughout the manufactured batches till date.

3.6.2 CHANGES IN FORMULATION & PROCESS

No change is made in the formulation as well as process in regular practice. It has been observed that every batch manufactured, following consistent formulation and process, is giving consistent results which are proven by the trend analysis for all the parameters covered and expressed in graphical form.

3.7 STABILITY MONITORING PROGRAMME

The results of the real-time and accelerated stability monitoring program of all the commercial batches carried out showed consistency of all Critical Quality Attributes (CQAs). The product's Critical Quality Attributes (CQAs) like pH, Assay, Appearance, Pyrogenicity and Sterility are all found to be within the set specifications in conformity with the limits of the pharmacopeia. The stability report is attached for reference as BML/QA/SBP/001 (2025)

3.8 QUALITY-RELATED RETURNS, COMPLAINTS AND RECALLS

3.8.1 MARKET COMPLAINTS

Up till now there is no market complaint received for this product during the period Jan. 2025 to Dec. 2025.

3.8.2 RETURNED GOODS FROM MARKET

No finished goods were returned from the market for the period Jan., 2025 to Dec., 2025 due to any quality problem

3.8.3 BATCH RECALLS

Up till now there is no batch recall for product during the period Jan., 2025 to Dec., 2025.

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3.8.4 SALVAGED GOODS

There is no salvaging of goods for the period of Jan., 2025 to Dec., 2025.

3.9 CORRECTIVE ACTIONS

No critical, major or minor deviations have been observed during the year under review. As such, no noteworthy corrective action was necessary. Few observed deviations during the year under review fall under the category of Others, which has to do with colour or print on tertiary packaging.

3.10 VALIDATED ANALYTICAL AND MANUFACTURING PROCEDURES

MANUFACTURING PROCEDURE

Equipment cleaning and sterilization
Dispensing of the raw material
Solution preparation
Batch coding of PVC bag (packaging material)
Solution filling
Sterilization of the filled solution in qualified autoclave
Final packaging of the sterilized solutions

ANALYTICAL PROCEDURE

In-process testing of the cleanliness of equipment
In-process chemical testing of the semi-finished products for critical quality attributes like pH, assay
Chemical testing of the finished products
Microbiological tests on the finished product-
Sterility tests
Bacterial endotoxin tests (Pyrogen and Limulus Amoebocyte Lysate)

SUMMARY FOR PROCESS VALIDATION:

We have manufactured 210 batches of Dextrose 5% Saline Intravenous Injection 500ml with same manufacturing formula since Jan., 2025 till Dec., 2025 and yearly APQR data shows satisfactory trend throughout the period. Also the retained samples prove that the product manufactured with the existing formulation is stable.

ENVIRONMENTAL CONDITIONS DURING MANUFACTURING OPERATION

The data of environmental conditions during the manufacturing of all batches were reviewed and recorded in the APQR. The environmental conditions of the critical areas were properly monitored in the raw material store, mixing unit, filling unit, batch coding unit and the storage areas.

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Environmental Conditions
Std Temp. : NMT 30°C
Relative Humidity: NMT 70%

CRITICAL EQUIPMENT PERFORMANCE

The critical equipment used during the manufacturing and analysis of the product were identified and overall performance of these equipment were evaluated by reviewing the breakdown and preventative maintenance records.

3.11 REVIEW OF OBSERVED ADVERSE DRUG REACTIONS

No adverse drug reaction (ADR) was observed with the product during the year under review.

3.12 REVIEW OF PERIODIC RISK-BENEFIT EVALUATION REPORTS

Dextrose 4.3% Saline Intravenous Injection is an intravenous fluid and electrolyte replenishment preparation officially approved for sale in Nigeria since the year 2020. The periodic safety update reports submitted for the product covering January–December 2025 indicated a favourable risk-benefit profile, with no adverse risk signals identified during the review period. All columns under the spontaneous safety signal and risk report evaluation for the System Organ Class (SOC) returned zero entries. About 640,000 bags of Bioflex Dextrose 4.3% Saline Intravenous Injection were manufactured and distributed across the nation in the year under review, and no report indicating either serious or non-serious early signal detection and/or safety risks on any essential System Organ Class was recorded.

3.13 REVIEW OF POST MARKETING SURVEILLANCE

The reports of clinical and therapeutic performance of the products from major health institutions across the nation have proven the product to be safe and efficacious. No adverse reaction has been reported, no report of product contamination nor pyrogenicity has been reported with the product within the year under review.

3.14 COMPARATIVE PRODUCT QUALITY REVIEW

CONCLUSION

All evaluated CQAs demonstrated that process control limits are within the specified limits at both intermediate and finished product stages, where applicable.

Process capability indices (Cpk) for pH, sodium chloride content, and dextrose content indicate acceptable to good process performance (Cpk > 1.0), confirming that the manufacturing process is capable and under statistical control. Rinse water analysis at the intermediate stage also showed adequate process

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capability. HMF was only evaluated at the finished product stage and demonstrated a high capability index (Cpk = 6.68), indicating excellent process control.

Overall, the manufacturing process is stable, capable, and consistently produces products that meet predefined quality specifications.

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